

Shopping Customer Analytics Dashboard Using MySQL and Power BI

1. Introduction

In today's digital retail environment, businesses generate large amounts of customer and sales data every day. Analyzing this data helps organizations understand customer behaviour, improve marketing strategies, optimize inventory, and increase overall revenue. Business Intelligence (BI) tools and data analytics techniques play a major role in transforming raw data into meaningful insights.

This project focuses on developing a Shopping Customer Analytics Dashboard using MySQL and Power BI. The dataset contains customer purchase details such as product category, purchase amount, subscription status, payment method, age group, promo code usage, shipping type, review rating, and previous purchases.

The main objective of the project is to analyse customer purchasing behaviour and visualize important business insights through an interactive dashboard. SQL queries are used for data analysis and preprocessing, while Power BI is used to create visualizations, KPIs, cards, filters, and charts.

The dashboard enables businesses to identify top-selling products, high-revenue categories, customer trends, subscription behaviour, and the effect of discounts and promo codes on purchases.

2. Problem Statement

Retail businesses often collect customer data but fail to utilize it effectively for decision-making. Without proper analysis, it becomes difficult to identify customer preferences, revenue-driving products, and purchasing trends.

The main problems addressed in this project are:

- Identifying the most purchased products within each category.
- Understanding customer purchasing behaviour based on age groups.
- Measuring revenue contribution of different customer segments.
- Analyzing whether repeat buyers are more likely to subscribe.
- Understanding the impact of discounts and promo codes on sales.
- Creating an interactive dashboard for better business decision-making.

The project aims to solve these problems by combining SQL-based data analysis with Power BI dashboard visualization.

3. Objectives of the Project

The major objectives of this project are:

1. To analyze customer shopping behaviour using SQL queries.
 2. To identify top-performing products and categories.
 3. To visualize revenue contribution by different customer groups.
 4. To study the relationship between subscriptions and repeat purchases.
 5. To build an interactive dashboard using Power BI.
 6. To generate business insights for strategic decision-making.
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4. Dataset Description

The dataset used in this project contains customer shopping records. Each row represents a customer purchase transaction.

Dataset Attributes

```
<class 'pandas.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                          3900 non-null   int64
1   Age                                  3900 non-null   int64
2   Gender                              3900 non-null   str
3   Item Purchased                       3900 non-null   str
4   Category                            3900 non-null   str
5   Purchase Amount (USD)                3900 non-null   int64
6   Location                             3900 non-null   str
7   Size                                 3900 non-null   str
8   Color                                3900 non-null   str
9   Season                               3900 non-null   str
10  Review Rating                        3863 non-null   float64
11  Subscription Status                  3900 non-null   str
12  Shipping Type                       3900 non-null   str
13  Discount Applied                    3900 non-null   str
14  Promo Code Used                     3900 non-null   str
15  Previous Purchases                   3900 non-null   int64
16  Payment Method                      3900 non-null   str
17  Frequency of Purchases               3900 non-null   str
dtypes: float64(1), int64(4), str(13)
memory usage: 548.6 KB
```

5. Technologies Used

5.1 MySQL

MySQL is used for:

- Data storage
- Query execution
- Aggregation and filtering
- Ranking products
- Customer behaviour analysis SQL queries are used to calculate:
 - Revenue contribution
 - Top products
 - Subscription rate
 - Purchase frequency
 - Repeat buyer analysis

5.2 Power BI

Power BI is used for:

- Interactive dashboard creation
- Data visualization
- KPI cards
- Bar charts and pie charts
- Filters and slicers
- Business insight representation

5.3 DAX (Data Analysis Expressions)

DAX measures are used in Power BI for:

- Dynamic calculations
 - Top product identification
 - Revenue measures
 - Ranking calculations
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6. Methodology

The project follows a structured analytical approach:

Step 1: Data Collection

Customer shopping data was collected and imported into MySQL.

Step 2: Data Cleaning

The dataset was checked for:

- Missing values
- Duplicate records
- Inconsistent data
- Incorrect formatting

Step 3: SQL Data Analysis

SQL queries were written to:

- Count purchases
- Calculate total revenue
- Rank products
- Analyze subscriptions
- Find top categories
- Measure promo code usage

Step 4: Data Import to Power BI

The processed dataset and query results were imported into Power BI.

Step 5: Dashboard Development

Interactive visuals and KPIs were created using Power BI.

Step 6: Insight Generation

Business insights were generated from dashboard analysis.

7. SQL Queries and Business Questions Used

The following SQL queries were used to analyze customer shopping behaviour and generate insights for the Power BI dashboard.

Q1. What is the total revenue generated by male vs female customers

```
select gender, sum(purchase_amount) as total_revenue
from customers
group by gender;
```

Purpose

This query compares revenue generated by male and female customers.

Q2. Which customers used a discount but still spent more than the average purchase amount?

```
select customer_id, purchase_amount -- count(customer_id) as 'total_id'
from customers
where promo_code_used = 'Yes' and purchase_amount > (select avg(purchase_amount) from customers) ;
```

Purpose

This query identifies high-value customers who used promotional discounts.

Q3. Which are the top 5 products with the highest average review rating?

```
select item_purchased, round(avg(review_rating),2) as 'avg_rating'
from customers
group by item_purchased
order by avg(review_rating) desc
limit 5;
```

Purpose

This query helps identify highly rated products based on customer reviews.

Q4. Compare the average purchase amounts between Standard and Express Shipping.

```
select shipping_type, round(avg(purchase_amount),2) as 'avg_purchase_amount'
from customers
where shipping_type in ('Express','Standard')-- shipping_type = 'Express' or shipping_type = 'Standard'
group by shipping_type;
```

Purpose

This query compares customer spending based on shipping methods.

Q5. Do subscribed customers spend more?

```
select subscription_status, count(customer_id) as 'total_customer',
round(avg(purchase_amount),2) as 'avg_purchase_spend',
round(sum(purchase_amount),2) as 'total_revenue'
from customers
group by subscription_status
order by total_revenue, avg_purchase_spend desc;
```

Purpose

This query analyzes whether subscription customers contribute more revenue.

Q6. Which 5 products have the highest percentage of purchases with discounts applied?

```
select item_purchased , count(item_purchased) as 'total_item',
round(100 * sum( case when promo_code_used = 'Yes' then 1 else 0 end) / count(*) ,2) as 'discount_rate'
from customers
group by item_purchased
order by discount_rate desc
limit 5;
```

Purpose

This query identifies products most frequently purchased using promo codes.

Q7. Segment customers into New, Returning, and Loyal customers.

```
with customer_type as (  
    select customer_id, previous_purchases,  
    case  
        when previous_purchases = 1 then 'new'  
        when previous_purchases between 2 and 10 then 'Returning'  
        else 'Loyal'  
    end as 'customer_segment' from customers )  
  
select customer_segment , count(*) as 'number_of_customers'  
from customer_type  
group by customer_segment;
```

Purpose

This query categorizes customers based on their purchase history.

Q8. What are the top 3 most purchased products within each category?

```
WITH item_count AS (  
    SELECT  
        category,  
        item_purchased,  
        COUNT(customer_id) AS total_customers  
    FROM customers  
    GROUP BY category, item_purchased  
)  
  
SELECT *  
FROM (  
    SELECT *,  
        ROW_NUMBER() OVER (  
            PARTITION BY category  
            ORDER BY total_customers DESC  
        ) AS item_rank  
    FROM item_count  
)  
ranked  
WHERE item_rank <= 3;
```

Purpose

This query ranks products within each category based on purchase count.

Q9. Are repeat buyers also likely to subscribe?

```
SELECT
    CASE
        WHEN previous_purchases > 5
        THEN 'Repeat Buyer'
        ELSE 'Non Repeat Buyer'
    END AS buyer_type,

    COUNT(*) AS total_customers,

    SUM(
        CASE
            WHEN subscription_status = 'Yes'
            THEN 1
            ELSE 0
        END
    ) AS subscribed_customers,

    ROUND(
        100 * SUM(
            CASE
                WHEN subscription_status = 'Yes'
                THEN 1
                ELSE 0
            END
        ) / COUNT(*),
        2
    ) AS subscription_rate

FROM customers

GROUP BY buyer_type;
```

This query measures the relationship between repeat purchases and subscriptions.

Q10. What is the revenue contribution of each age group?

```
SELECT
    age_group,
    SUM(purchase_amount) AS total_revenue,
    ROUND(
        100 * SUM(purchase_amount) /
        (SELECT SUM(purchase_amount) FROM customers),
        2
    ) AS revenue_contribution_percentage
FROM customers
GROUP BY age_group
ORDER BY revenue_contribution_percentage DESC;
```

Purpose

This query identifies which customer age groups contribute the most revenue.

8. Dashboard Visualizations

The dashboard contains multiple visuals and KPIs for better analysis.

KPI Cards

- Total Revenue
- Total Customers
- Average Purchase Amount
- Top Product by Category

Charts Used

Bar Chart

Used for:

- Revenue by category
- Top purchased products **Pie/Donut Chart**

Used for:

- Revenue contribution by age group
- Subscription analysis **Card Visuals**

Used for:

- Top products
- Revenue metrics
- Customer count

Slicers

Used for filtering data based on:

- Category
- Gender
- Season
- Subscription status

9. Key Insights

The dashboard analysis generated several important business insights:

1. Certain product categories contribute significantly higher revenue than others.
 2. Adult customers generate the highest revenue contribution.
 3. Repeat buyers are more likely to subscribe compared to new customers.
 4. Promo codes and discounts influence customer purchasing behaviour.
 5. Some products dominate sales within their categories.
 6. Subscription customers tend to purchase more frequently.
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Dashboard view



10. Advantages of the Project

- Helps businesses understand customer behaviour.
- Provides interactive and dynamic reporting.
- Improves business decision-making.
- Identifies high-performing products.
- Helps analyze marketing effectiveness.
- Enhances customer segmentation analysis.
- Easy to interpret through visualization.

11. Future Scope

The project can be enhanced further by:

- Integrating machine learning for sales prediction.
- Performing sentiment analysis using customer reviews.
- Developing real-time dashboards.
- Adding customer recommendation systems.
- Integrating cloud databases and live APIs.
- Creating mobile-friendly BI dashboards.

12. Conclusion

The Shopping Customer Analytics Dashboard successfully demonstrates how SQL and Power BI can be combined to analyze customer shopping behaviour and generate meaningful business insights.

MySQL was used for efficient data querying and analytical processing, while Power BI enabled interactive visualization and dashboard creation. The project helps identify top-performing products, customer purchasing patterns, revenue-driving categories, and subscription trends.

The dashboard provides a practical solution for retail businesses to make data-driven decisions and improve overall business performance. This project also demonstrates the importance of Business Intelligence tools in modern analytics and decision-making systems.

13. References

1. Microsoft Power BI Documentation
2. MySQL Official Documentation
3. SQL Window Functions Documentation
4. Business Intelligence and Data Analytics Study Materials